Instrumentation & Control System Engineer

Muhammad Hassan

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SUMMARY

Electrical Engineering graduate (FAST-NU, 2026) with hands-on field experience from NTDC internship on live 220kV transmission systems, including safety-critical PTW-to-re-energization procedures. Completed intensive PLC/HMI/SCADA training (Siemens TIA Portal) with full hardware-software implementation experience at DTI. Willingness to learn and apply automation and control systems expertise to real-world industrial process and power systems. Strong analytical and problem-solving skills with a proven ability in technical documentation, reporting, and cross-functional team coordination.

AREAS OF EXPERTISE

PLC Programming: Siemens TIA Portal (S7-1200/1500), Ladder Logic, Familiarity with Delta, Allen-Bradley, Mitsubishi

HMI/SCADA: Siemens HMI & WinCC Design, Weintek HMI Design, Profinet HMI-PLC Integration, Real-time Tag Monitoring & Alarming

Industrial Communication: Modbus, Profinet (Ethernet), Profibus, RS232/RS485

Instrumentation: Analog/Digital I/O, VFDs, Sensors (NPN/PNP)

Power Systems: 220kV Transmission, SCADA, Protection Relay Coordination, SLD Reading

Control Theory: Process Control Logic, SIL Fundamentals, Basic P&IDs and DCS Concepts

Panel Design & Wiring: Control Panel Wiring, AutoCAD Electrical (Basics), VFD/Motor Wiring, and Hardware Circuit Troubleshooting

TRAINING & CERTIFICATIONS

PLC / HMI / SCADA Training, DTI, Lahore (Siemens TIA Portal, S7-1200 & S7-1500) *Jul – August 2026*

- Hands-on PLC programming (Siemens) and HMI/SCADA fundamentals; theoretical coverage of Allen-Bradley, Mitsubishi, Delta, Omron; industrial protocols (Modbus, Profinet(Ethernet), Profibus, RS232/RS485); digital/analog I/O, VFDs, P&IDs, servo/stepper drives, motor control circuits, panel wiring, and NPN/PNP sensor circuits.

Electrical Engineering Internship Certificate, NTDC (National Transmission & Despatch Co.) *Jul – Aug 2025*

- Completed a hands-on internship covering live 220 kV transmission systems, SCADA-based monitoring, and protection relay coordination.

WORK EXPERIENCE

Electrical Engineering Intern, NTDC (National Transmission & Despatch Company) *Jul 7 – Aug 29, 2025*

- Supported real-time SCADA-based monitoring of voltage and load parameters with senior engineers, gaining direct exposure to NOC-style monitoring and control operations under strict safety protocols.
- Worked on live 220 kV transmission systems and participated in a full PTW-to-re-energization field maintenance job — isolated a 250 MVA transformer from the GIS hall, discharged the PT, diagnosed a faulty trip coil via SLD verification, corrected the fault, and supervised load ramp-up on re-energization.
- Authored a structured technical report on operational procedures and equipment, adopted by the team as a post-internship reference resource.

RELEVANT PROJECTS

Conveyor Belt Material Sorting System (Metal/Plastic Detection with Hardware) *2026*

- Built and programmed a sensor-based material classification system on a conveyor line, using proximity sensors to differentiate metal and non-metal (plastic) parts and route them via actuator-driven sorting; implemented full hardware-side sorting logic and sequencing.

3-Phase VFD Motor Speed Control (iKon 300 Series Hardware) *2026*

- Programmed a 3-phase VFD to control induction motor speed via frequency, using keypad-based parameter setup for jogging, direction control, and soft-start/stop.

Traffic Light Control with Dynamic Pedestrian Timer Extension (Siemens S7-1200 + WinCC HMI) *2026*

- Programmed a 4-way intersection light sequence using TP/TON/TOF timers, with pedestrian-triggered logic that extends the running green-light timer in real time without resetting the cycle.

Digital Twin – Battery Monitoring System *2026*

- Developed an embedded digital twin for real-time battery voltage, current, and temperature monitoring, using an ML model to detect anomalies and trigger automated protective actions (cell isolation, smart load activation).

EDUCATION

B.S. Electrical Engineering — FAST-NU (National University of Computer & Emerging Sciences) *Completed Jun 2026*

F.Sc. Pre-Engineering — KIPS College, Lahore *Completed Sep 2022*

LEADERSHIP

Deputy Head, Softec FAST-NU (2022–2025), coordinated logistics and operations for one of Pakistan's largest student-run tech competitions.

LANGUAGES

English (Fluent) | Urdu (Native) | Punjabi (Native)